NUMERICAL METHODS AND INTEGRAL TRANSFORMS

I B.TECH - II SEMESTER								
Course Code	**Category**	**Hours/Week**			**Credits**	**Maximum Marks**		
A6BS02	BSC	**L**	**T**	**P**	**C**	**CIE**	**SEE**	**Total**
		3	1	-	4	40	60	100
Contact Classes: 44	**Tutorial Classes: 08**	**Practical Classes: Nil**				**Total Classes: 52**		

COURSE OBJECTIVES

The course will enable the students to:

1. Curve fitting and Interpolation techniques.
2. Numerical techniques.
3. Fourier series for periodic function.
4. Laplace transforms.
5. Concept and application of Fourier Transforms and Vector differentiation.

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Apply Curve fitting and Interpolation techniques.
2. Apply various numerical techniques.
3. Find the Fourier series of the periodic functions.
4. Obtain the Laplace transforms of functions.
5. Find Fourier transforms and apply vector differentiation techniques.

UNIT - I	INTERPOLATION AND CURVE FITTING	CLASSES: 08

INTERPOLATION: Finite differences: Forward, Backward and Central differences - Other difference operators and relations between them - Difference of a polynomial – Missing terms - Newton's forward interpolation, Newton's backward interpolation, Gauss's forward and backward interpolation formulae. Interpolation with unequal intervals – Lagrange's interpolation.
CURVE FITTING: Method of least squares - Fitting a straight line, second degree parabola and non-linear curves of the form $y= a e^{bx}$, $y= a x^{b}$, $y = a b^{x}$ by the method of least squares.

UNIT - II	NUMERICAL TECHNIQUES	CLASSES: 08

ROOT FINDING TECHNIQUES:
Bisection method Regulafalsi method, Iteration method and Newton Raphson method.
NUMERICAL INTEGRATION :
Trapezoidal rule - Simpson's one-third rule - Simpson's three-eighth rule.
NUMERICAL SOLUTION OF ORDINARY DIFFERENTIAL EQUATIONS:
Taylor's series method – Euler's - modified Euler's Method – Runge-Kutta method.

UNIT - III	FOURIER SERIES	CLASSES: 10

Periodic function-Determination of Fourier Coefficients-Fourier Series-Even and Odd functions-Fourier series in arbitrary interval-Half range Fourier sine and cosine expansions.

UNIT - IV	LAPLACE TRANSFORMS	CLASSES: 10

Laplace transforms of elementary functions- First shifting theorem - Change of scale property – Multiplication by t^n- Division by t – Laplace transforms of derivatives and integrals – Unit step function – Second shifting theorem – Periodic function – Evaluation of integrals by Laplace transforms – Inverse Laplace transforms-Method of partial fractions – Other methods of finding inverse transforms – Convolution theorem – Applications of Laplace transforms to ordinary differential equations.

UNIT - V	FOURIER TRANSFORMS AND VECTOR DIFERENTIATION	CLASSES: 10

Fourier Integral theorem (Statement only)-Fourier Sine and Cosine Integrals, Fourier Transforms, Cosine and Sine transforms, properties, Inverse transforms.
Vector functions, vector differentiation, gradient, directional derivative, divergence, curl and scalar potential.

TEXT BOOKS

1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
2. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.

REFERENCE BOOKS

1. G.B.Thomas, calculus and analytical geometry,9th Edition, Pearson Reprint 2006.
2. N.P Bali and Manish Goyal ,A Text of Engineering Mathematics, Laxmi publications,2008.
3. E.L.Ince, Ordinary differential Equations, Dover publications,1958.

WEB REFERENCES

1. https://www.efunda.com/math/math_home/math.cfm
2. https://www.ocw.mit.edu/resources/#Mathematics
3. https://www.sosmath.com/
4. https://www.mathworld.wolfram.com/

E-TEXT BOOKS

1. https://www.e-booksdirectory.com/details.php?ebook=10166

MOOCS COURSE

1. https://swayam.gov.in/
2. https://onlinecourses.nptel.ac.in/